
SMART SKINCARE RECOMMENDER

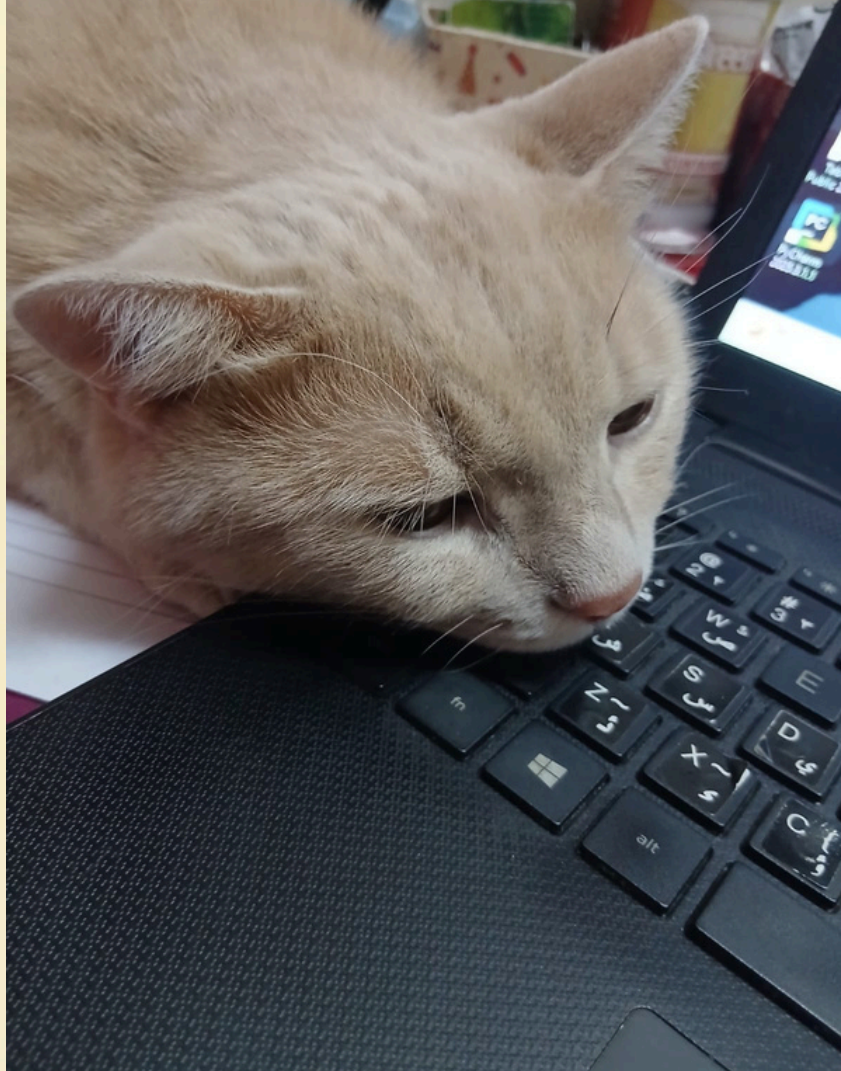
Presented By : Omnia Emad



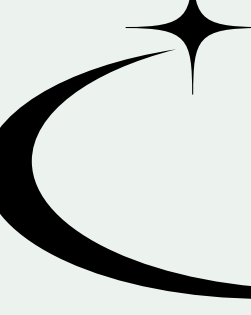
PROBLEM STATEMENT

- **Hard to Choose:** Finding the right skincare product is overwhelming due to endless options and confusing ingredient lists.
- **Risky Decisions:** Using the wrong active ingredients for specific skin issues (like acne or eczema) can cause irritation or make the skin worse.
- **No Quick Guidance:** People often lack fast, accessible tools to analyze their skin conditions and filter trustworthy product reviews accurately.



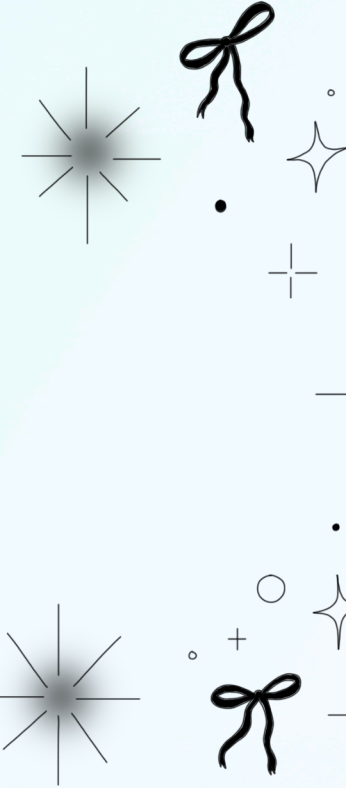


TEAM



- Omnia Emad
- Cookies
- Gemini

SYS ARCHITECTURE



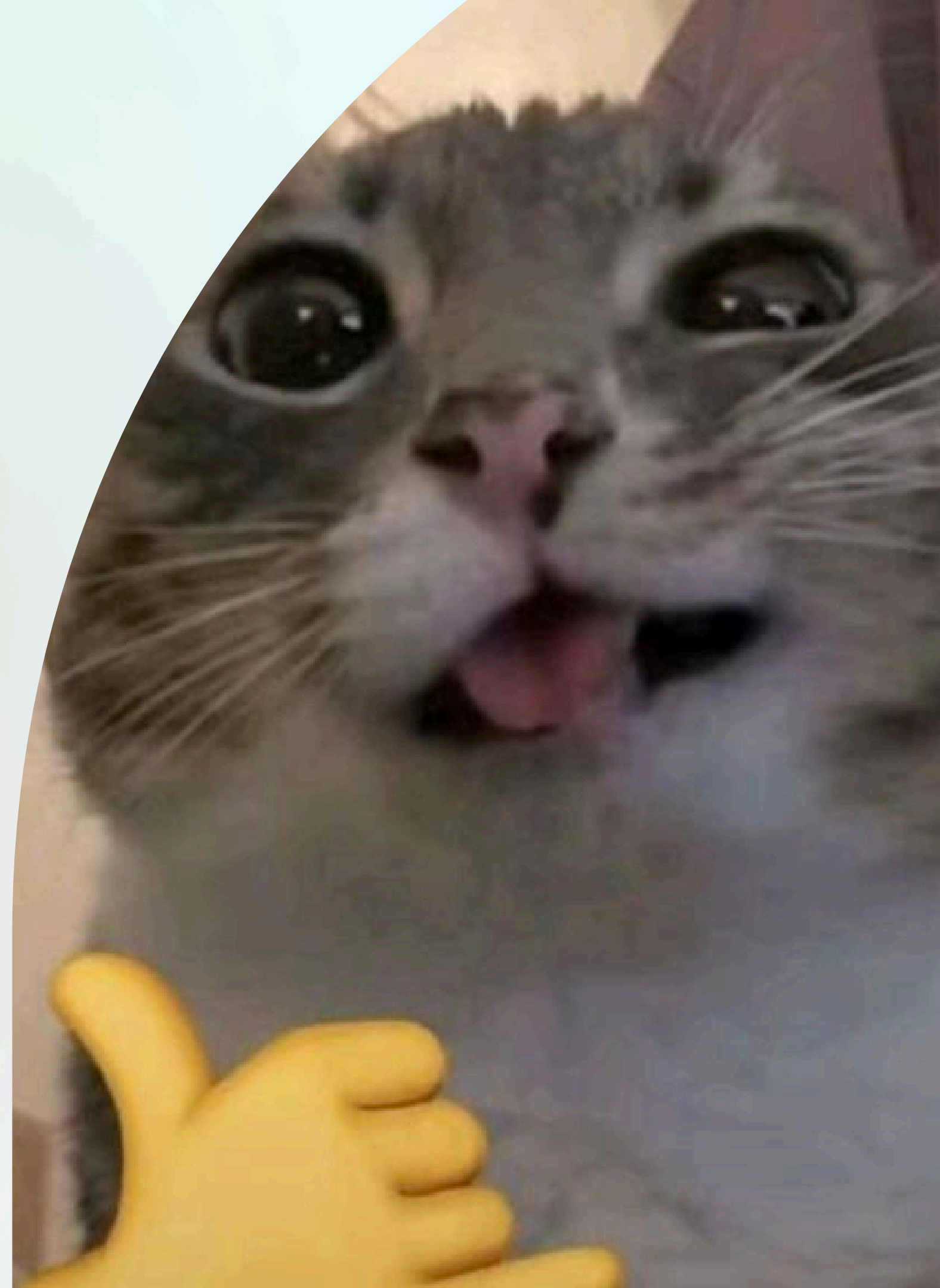
- **Input:** User uploads a skin image & profile data.
- **Computer Vision:** Analyzes the photo to detect the skin condition.
- **Recommender Engine:** Matches active ingredients to treat the issue safely.
- **NLP Sentiment Analysis:** Analyzes product reviews to score overall customer satisfaction.
- **Output:** Displays tailored skincare recommendations on the Streamlit dashboard.

FIRST: COMPUTER VISION

Skin Problems

- blackhead
- comedonical-acne
- eczema
- pustule-acne
- rosacea
- wrinkle

Data Roboflow [link](#)



RECOMMENDATION ENGINE (RULE-BASED)

-  Primary Actives: Targeted ingredients mapped directly to treat specific skin conditions (e.g., Salicylic Acid for Acne, Retinol for Wrinkles).
-  Supportive Ingredients: Beneficial compounds that enhance product performance and maintain skin hydration (e.g., Glycerin, Hyaluronic Acid).
-  Irritants Safety Filter: Flags potential allergens or sensitive-skin triggers (e.g., Fragrance, Alcohol Denat) to protect the user's skin barrier.

PRODUCT SENTIMENT ANALYSIS (NLP)

- **Review Processing:** Analyzes real user feedback from product reviews using Natural Language Processing.
- **Satisfaction Scoring:** Calculates the ratio of positive to negative reviews for each recommended product.
- **Trust & Transparency:** Helps users make confident decisions based on real community experiences alongside scientific active ingredients.



DEMO BY STREAMLIT

Now, let's move to the
interactive demo to see how
our end-to-end system
works in real time



LINKS

- Sephora Datasets [Kaggle](#)
- Sephora Cleaning & EDA [Colab](#)
- Skin Detection [Colab](#)
- Recommendation Engine [Colab](#)
- NLP Sentiment Analysis [Colab](#)



A close-up, slightly blurred photograph of a baby lying on its back. The baby is wrapped in a thick, red, textured blanket. The baby's face is visible at the top of the frame, looking upwards. The background is a light-colored, textured surface, possibly a carpet or rug. The text "THANK YOU" is overlaid in the center of the image in a white, serif font.

THANK YOU